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Modeling and identification of an electrostatic motor

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Abstract

The present study describes modeling and parameter identification for a high-power electrostatic motor. The motor is a synchronous motor with six phases that is modeled by a six-terminal capacitance network. A method to measure the capacitance coefficient matrix of a network using an LCR tester is proposed and demonstrated. The thrust force of the motor is shown analytically using the proposed model, which shows good agreement with empirical measurements of the motor behavior. Based on the proposed model, the effect of electrode skew is analyzed for force ripple reduction. Using the analysis, a skew motor is fabricated and tested, with the results demonstrating that a skewing electrode with a skew factor π can considerably reduce thrust force ripple.
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Keywords: Actuator; Electrostatic motor; Modeling; Parameter identification; Skew; Ripple reduction

1. Introduction

Since the late 1980s, several kinds of electrostatic motors have been studied and fabricated, mostly as MEMS actuators [1–5]. Electrostatic motors are considered to be one of the most promising MEMS actuators for two primary reasons: these motors have a planar structure that is highly compatible with IC lithography processes and the scaling law of electrostatic force states that the force per volume ratios of electrostatic actuators increase as their dimensions decrease, indicating that electrostatic motors could be stronger than electromagnetic actuators in the micro scale [1,2].

On the other hand, by integrating a number of micro actuator elements that have a large force/volume ratio, ordinary-sized actuators having a large force/volume ratio can also be obtained. Such integration can be easily realized in surface-drive type electrostatic motors. To date, several kinds of surface drive motors have been proposed [1,6–9]. Among these, the strongest is a linear synchronous motor called the *dual excitation multiphase electrostatic drive* (DEMED), which utilizes film electrodes for both stator and slider [8,9]. This motor has a characteristics length of centimeter-order and has demonstrated a thrust force of several tens of N and a moving speed of over 1 m/s. With such superior performance, electrostatic motors appear promising not only for micro machines but also ordinary-sized devices, such as positioning stages and robotic servo actuators.

For these practical applications, the motor should have a simple electrical model that can express the characteristics of the motor well and that has parameters that are easily identifiable. In the case of micro motors, the development of such models is difficult or impractical because the model parameters cannot be easily measured due to their extremely small size. However, with strong electrostatic motors, such as DEMED, the development of such models has become practical.

In the present paper, for a DEMED, we propose an electrical model for the motor and describe how to identify the model parameters. Based on the developed model, the thrust force characteristics are analyzed and improved.

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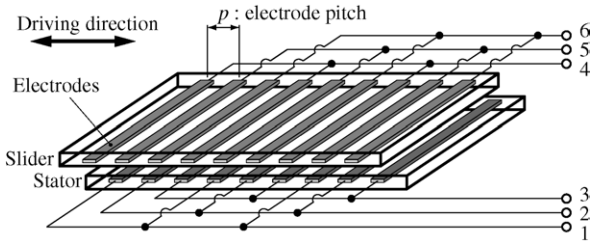


Fig. 1. Basic structure of DEMED.

2. Dual excitation multiphase electrostatic drive

2.1. Principle

The DEMED is a synchronous motor driven by three-phase ac voltages. Fig. 1 shows a schematic diagram of the basic DEMED, which consists of a pair of thin films that contain three-phase parallel electrodes that have the same electrode pitch p.

To drive the motor, the three-phase electrodes in the stator and the slider are excited by two sets of three-phase ac voltages. The excitation voltages can be expressed in a vector form as

$$\begin{aligned}\mathbf{v} &= [v_1, v_2, v_3, v_4, v_5, v_6]^{\mathrm{T}} \\ &= \left[v_{\mathrm{st}} \sin(\omega_{\mathrm{st}} t), v_{\mathrm{st}} \sin\left(\omega_{\mathrm{st}} t - \frac{2\pi}{3}\right), \right. \\ &\quad v_{\mathrm{st}} \sin\left(\omega_{\mathrm{st}} t + \frac{2\pi}{3}\right), v_{\mathrm{sl}} \sin(\omega_{\mathrm{sl}} t), \\ &\quad \left. v_{\mathrm{sl}} \sin\left(\omega_{\mathrm{sl}} t - \frac{2\pi}{3}\right), v_{\mathrm{sl}} \sin\left(\omega_{\mathrm{sl}} t + \frac{2\pi}{3}\right) \right]^{\mathrm{T}} \end{aligned} \quad (1)$$

where v_1 to v_6 represent the voltages on the electrodes of the corresponding numbers, and v_{st}, v_{sl}, ω_{st}, and ω_{sl} represent voltage amplitudes and angular frequencies of the two sets of three-phase ac voltages. The excitation generates running potential waves in the two films, as depicted in Fig. 2. Since the wavelengths of the waves are $3p$, the running speeds of the waves on the stator and the slider, u_{st} and $\hat{u}_{\mathrm{sl}}$, are

$$\begin{cases} u_{\mathrm{st}} = \dfrac{3p}{2\pi}\omega_{\mathrm{st}} \\ \hat{u}_{\mathrm{sl}} = \dfrac{3p}{2\pi}\omega_{\mathrm{sl}} \end{cases} \quad (2)$$

where the coordinate systems of u_{st} and $\hat{u}_{\mathrm{sl}}$ are fixed to the stator and the slider, respectively. Since they always have the same wavelength, the spatial phase difference δ between the waves can be defined. These waves interact to generate a tangential thrust force f_x that depends on the phase difference δ, as ideally depicted in Fig. 3. When the operation is synchronized under a constant load, the phase difference is preserved so that thrust force balances the load at stable equilibrium. That is, the slider wave becomes stationary relative to the stator wave. Since the coordinate system for $\hat{u}_{\mathrm{sl}}$ is moving relative to the u_{st} coordinate system with the slider speed u, slider speed u can be derived as

$$u = u_{\mathrm{st}} - \hat{u}_{\mathrm{sl}} = \frac{3p}{2\pi}(\omega_{\mathrm{st}} - \omega_{\mathrm{sl}}) \quad (3)$$

If ω_{sl} and v_{sl} are chosen so as to satisfy $\omega_{\mathrm{st}} t = -\omega_{\mathrm{sl}} t + \pi/3 \equiv \phi$, and $v_{\mathrm{st}} = v_{\mathrm{sl}} \equiv v_0$, i.e.

$$\begin{aligned}\mathbf{v} = v_0 \Bigg[&\sin(\phi), \sin\left(\phi - \frac{2\pi}{3}\right), \sin\left(\phi + \frac{2\pi}{3}\right), \\ &\sin\left(\phi + \frac{2\pi}{3}\right), \sin\left(\phi - \frac{2\pi}{3}\right), \sin(\phi) \Bigg]^{\mathrm{T}} \end{aligned} \quad (4)$$

then the system can be simplified as shown in Fig. 4. This type of operation is referred to as *single-frequency operation*, whereas the type of operation shown in Fig. 2 is referred to as *dual-frequency operation*. Practically speaking, due to its

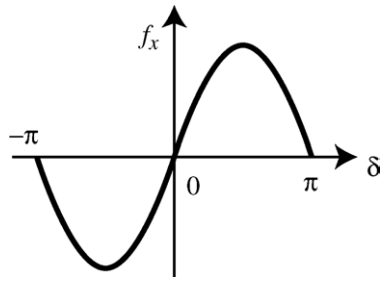


Fig. 3. Idealized relationship between thrust force f_x and spatial phase difference δ.

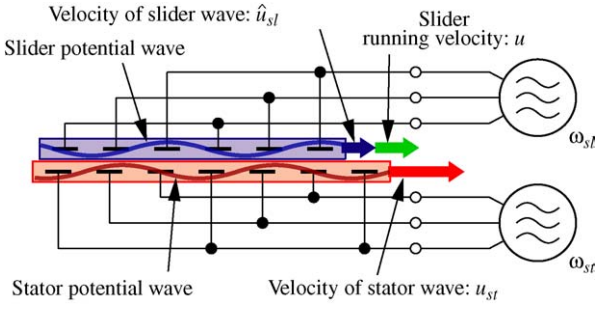


Fig. 2. Intersectional illustration describing the operation principle.

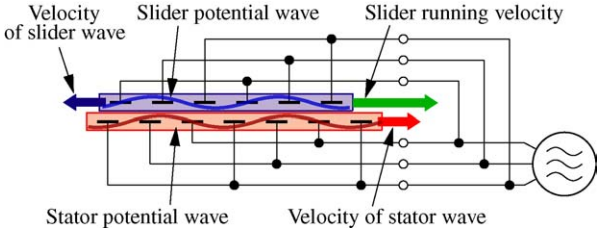


Fig. 4. Single-frequency operation.

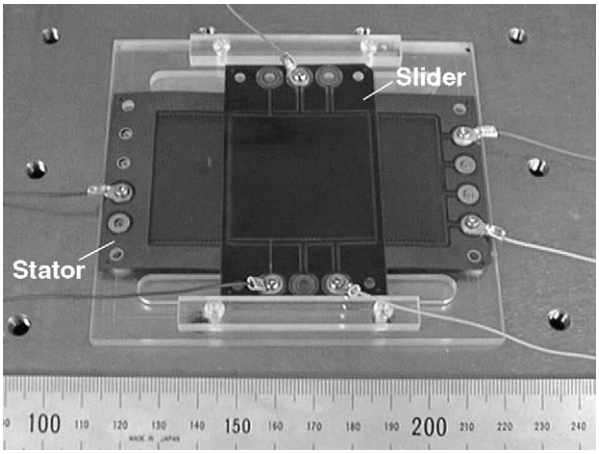


Fig. 5. Prototype motor.

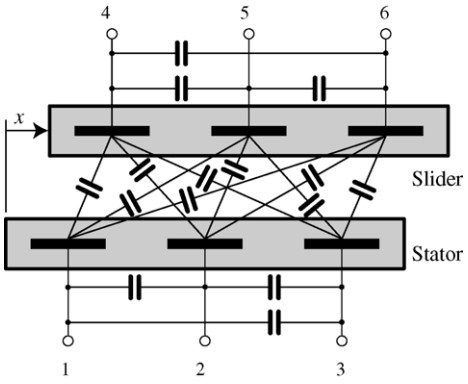


Fig. 6. Six-terminal capacitance network model.

simplicity, single-frequency operation is preferable to dual-frequency operation.

2.2. Prototype

Fig. 5 shows a prototype of the motor. The stator and the slider are fabricated using *flexible printed circuit* technology. The electrode pitch p of the prototype is 200 μm. The electrode area of the slider is measured as 80 mm × 40 mm, and that of the stator is measured as 40 mm × 40 mm. In this setup, the slider film is placed directly on the stator film. Since direct contact between the slider and the stator causes friction, glass beads of 20 μm in diameter are scattered between the films. As discussed later herein, the thrust force of the motor is proportional to the product of v_{st} and v_{sl}, or v_0^2 in single-frequency operation. Although the motor can work from around $v_0 = 500$ V, obtaining a practical force requires that v_0 should be higher than approximately 1 kV. With such a high voltage, the electrostatic field between the films exceeds the maximum field strength of air and a resultant discharge occurs in the air gap. Therefore, in order to prevent this discharge, dielectric liquid, *Fluorinert* FC-77 (3 M), is injected between the films for operation.

3. Motor model based on a capacitance matrix

3.1. Capacitance coefficient matrix

In the case of magnetic motors, the motor is electrically modeled using a serial *L–R* circuit. Considering the relationship between the magnetic motor and the electrostatic motor, the electrostatic motor should be modeled by a parallel *R–C* circuit [10]. The *R* components in these models represent the copper loss of the winding wires in the case of the magnetic motor and the dielectric loss of the insulating layers in the case of the electrostatic motor. Whereas magnetic motors have fairly large copper losses, because of their long winding wires, the dielectric loss in the electrostatic motor is usually small for the operating frequency range of up to 1 kHz, although the dielectric loss depends on the material used for the insulating layers. Therefore, in the present paper, we ignore the dielectric loss and provide a motor model with capacitances only.

Since DEMED has six phases in total, this motor can be modeled as a six-terminal capacitance network, of which six terminals are connected to each other by capacitances. Such a model is illustrated in Fig. 6. In multi-body electrostatic systems, the capacitances can be expressed using a matrix, which is referred to as a *capacitance coefficient matrix*, thus a motor can be represented by a single matrix. Provided that the ground plane is infinite and every electrostatic interaction is closed inside the motor, then the capacitance coefficient matrix **C** generally satisfies the following condition:

$$\mathbf{q} = \mathbf{C}\mathbf{v} \tag{5}$$

where **q** is a 1×6 vector that represents the charges on the six terminals. The matrix **C** has the following characteristics: (a) the matrix is symmetric; (b) the sum of each row or each column is zero; and (c) the diagonal elements are positive, whereas the other elements are negative.

3.2. Characteristics of the motor as derived from the capacitance coefficient matrix

One of the objectives of the present paper is to show that various characteristics of the motor can be derived from its capacitance matrix alone. This section provides simple equations that express the characteristics of the motor.

The force generated by a system described by matrix **C** can be derived by the virtual displacement approach; for systems consisting of conducting bodies with fixed voltages, the force f_x in the x direction can be calculated as the derivative of the total energy in terms of x. The total energy stored in a motor having capacitance matrix **C** is given as

$$U_v = \frac{1}{2}\mathbf{v}^{\mathrm{T}}\mathbf{C}\mathbf{v} \tag{6}$$

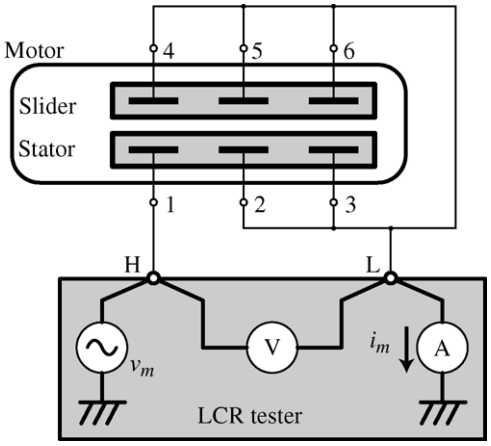


Fig. 7. Setup for measuring diagonal elements of **C**.

Therefore, the thrust force f_x for the motor is given as

$$f_x = \frac{\partial U_v}{\partial x} = \frac{1}{2}\mathbf{v}^{\mathrm{T}}\frac{\partial \mathbf{C}}{\partial x}\mathbf{v} \tag{7}$$

Using the matrix **C**, the relationship between voltage **v** and current **i**, which expresses the currents flowing into the six terminals, can be derived as

$$\mathbf{i} = \frac{\mathrm{d}\mathbf{q}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{C}\mathbf{v}) \tag{8}$$

Note that, when the motor is moving, both matrix **C** and voltage **v** change with time. When the motor is stationary, e.g. when fixed intentionally for some measurements, matrix **C** can be considered as a constant matrix, and Eq. (8) can be simplified to

$$\mathbf{i} = \mathbf{C}\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \tag{9}$$

3.3. Identification of the matrix

Methods by which to measure the capacitance coefficient matrix have been described [11]. However, although these methods can assure precise measurement, they require complicated procedures, and therefore are not practical for the identification of motor parameters. In this section, we propose a simple yet practical method, using a normal LCR tester, by which to determine the capacitance of the motor. Since the method cannot discriminate stray capacitances, it would not be applicable to micro motors. However, the method is effective for high-power motors for which the effect of stray capacitances is ignorable.

3.3.1. Measurement of diagonal elements

Fig. 7 shows the setup used to measure the diagonal elements of the matrix. In order to measure the ith diagonal element C_{ii}, only the ith terminal of the motor is connected to the "H" probe of the tester, and the other terminals are connected to the "L" probe. In order to describe the measurement principle, we define connection vector **R**, which describes the connection between the motor and the tester. Here, **R** is defined such that if the ith element of vector **R** is 1, then the ith terminal is connected to the "H" probe of the tester, and if the ith element is 0, then the ith terminal is connected to the "L" probe. For example, in case of the setup illustrated in Fig. 7, we have $\mathbf{R} = [1, 0, 0, 0, 0, 0]$.

Typically, an LCR tester applies voltage v_{m} to the "H" probe, and measures current i_{m} flowing through the "L" probe. Then, the impedance is given by the relation between v_{m} and i_{m}. In the connection shown in Fig. 7, the voltage vector **v** and current vector **i** of the motor satisfy the following equations:

$$\mathbf{v} = v_{\mathrm{m}}\mathbf{R}^{\mathrm{T}} \tag{10}$$

and

$$i_{\mathrm{m}} = \mathbf{R}\mathbf{i} \tag{11}$$

Using the above two equations and Eq. (9), the following relationship can be obtained:

$$\begin{aligned} i_{\mathrm{m}} &= \mathbf{R}\mathbf{i} \\ &= \mathbf{R}\mathbf{C}\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \\ &= \frac{\mathrm{d}v_{\mathrm{m}}}{\mathrm{d}t}\mathbf{R}\mathbf{C}\mathbf{R}^{\mathrm{T}} \end{aligned} \tag{12}$$

Since an LCR tester basically gives capacitance as the ratio between i_{m} and $\mathrm{d}v_{\mathrm{m}}/\mathrm{d}t$, the measured capacitance C_{meas} becomes

$$C_{\mathrm{meas}} = \mathbf{R}\mathbf{C}\mathbf{R}^{\mathrm{T}} \tag{13}$$

If **R** is $[1, 0, 0, 0, 0, 0]^{\mathrm{T}}$, then Eq. (13) gives $C_{\mathrm{meas}} = C_{11}$. In such a manner, the diagonal element C_{ii} can be measured directly by the setting shown in Fig. 7.

3.3.2. Measurement of non-diagonal elements

Fig. 8 shows the setting for the measurement of non-diagonal elements. In order to measure C_{ij}, both the ith and jth terminals are connected to the "H" probe. In the case of

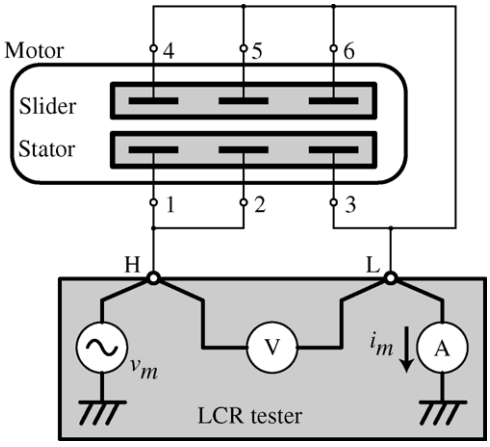


Fig. 8. Setup for measuring non-diagonal elements of **C**.

Fig. 8, the first and second terminals are connected to "H", and therefore we have $\mathbf{R} = [1, 1, 0, 0, 0, 0]^{\mathrm{T}}$. From Eq. (13), the measured capacitance is given as

$$\begin{aligned} C_{\mathrm{meas}} &= C_{11} + C_{12} + C_{21} + C_{22} \\ &= C_{11} + 2C_{12} + C_{22} \end{aligned} \tag{14}$$

Note that $\mathbf{C}$ is symmetric. In the case of the non-diagonal elements, the capacitance cannot be directly measured. However, from Eq. (14), non-diagonal element C_{12} can be easily derived because diagonal elements C_{11} and C_{22} are already known, as described previously.

4. Parameter identification of the prototype DEMED

4.1. Measurement

The capacitance coefficient matrix for DEMED is measured using the proposed method for the prototype motor shown in Fig. 5. The measured results are shown in Fig. 9. Since capacitances repeat every 600 μm, which is the same as $3p$, we define the electric angle θ_x of position as

$$\theta_x = \frac{2\pi}{3p}x \tag{15}$$

In Fig. 9, considering that the three phases of the motor are symmetric in principle, elements that should have the same values are grouped. However, some offsets can be observed among the values measured for the same groups. In the fabricated electrode films, three-phase electrodes are connected into three feeding electrodes located at the edges of the films, as shown in Fig. 10. The three feeding electrodes cannot be made symmetric because of the geometrical limitation, causing an imbalance in the measured capacitances. However, since the feeding electrodes do not contribute to thrust force generation, these offsets can be ignored.

4.2. Approximation of the capacitance coefficient matrix for a mathematical representation

The measured capacitances can be approximated by sine functions. Fig. 9 indicates that measured capacitances contain up to third-order harmonics. Therefore, ignoring the offset errors, the capacitances in Fig. 9 can be approximated as:

$$\begin{cases} C_{11}(= C_{22} = C_{33}) = C_{st0} + C_{st3}\cos(3\theta_x) \\ C_{44}(= C_{55} = C_{66}) = C_{sl0} + C_{sl3}\cos(3\theta_x) \\ C_{12}(= C_{13} = C_{23}) = -C_{t0} - C_{t3}\cos(3\theta_x) \\ C_{45}(= C_{46} = C_{56}) = -C_{l0} - C_{l3}\cos(3\theta_x) \\ C_{14}(= C_{25} = C_{36}) = C_m(\theta_x) \\ C_{15}(= C_{26} = C_{34}) = C_m(\theta_x + 2\pi/3) \\ C_{16}(= C_{24} = C_{35}) = C_m(\theta_x - 2\pi/3) \end{cases} \tag{16}$$

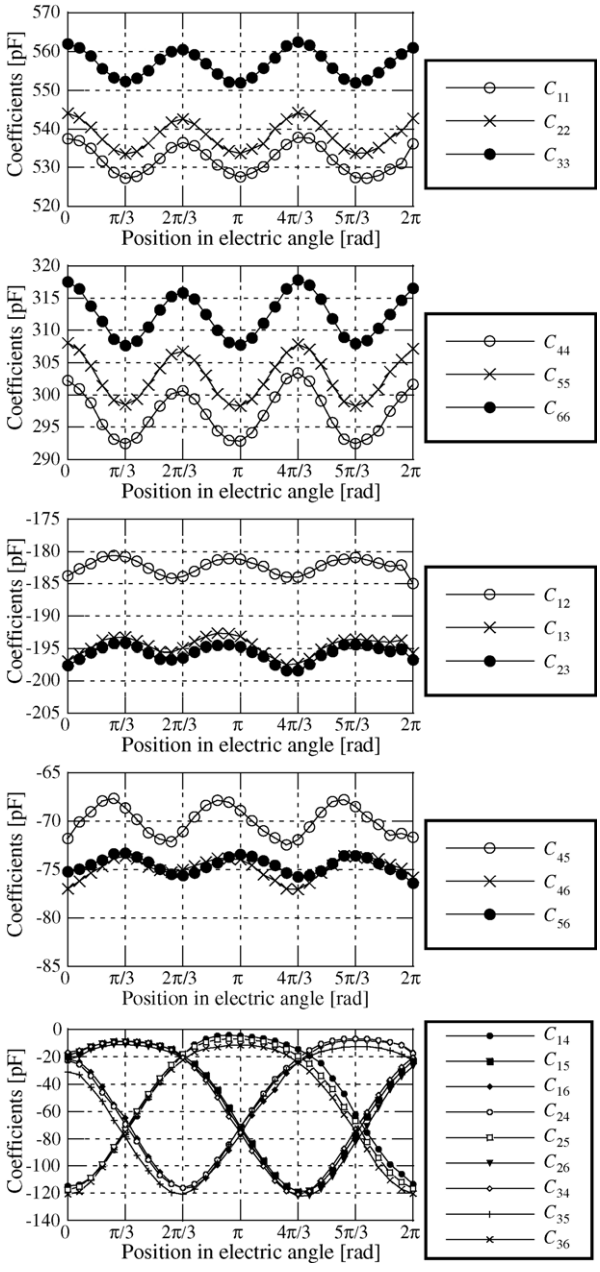


Fig. 9. Measured capacitance coefficient matrix for the prototype motor.

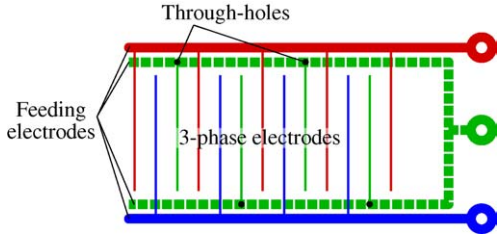


Fig. 10. Imbalance of three-phase electrodes in the feeding lines. The electrode illustrated by dashed lines is located on the back-face layer, whereas the others are located on the front-face layer.

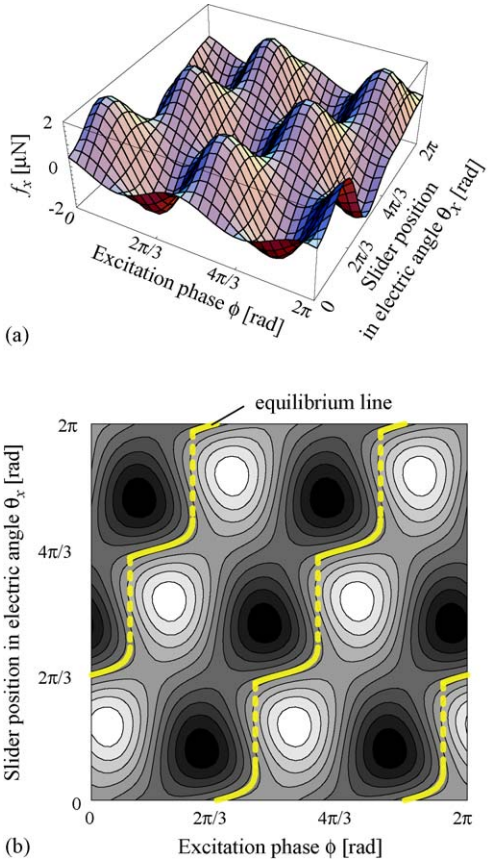


Fig. 11. Thrust force characteristics derived from the capacitance matrix. Voltage amplitude v_0 is normalized to 1 V. (a) Three-dimensional plot and (b) contour plot with equilibrium line.

where $C_m(\theta_x) = -C_{m0} - C_{m1}\cos(\theta_x) - C_{m2}\cos(2\theta_x) - C_{m3}\cos(3\theta_x)$. By substituting the measured values into Eq. (16), the parameters are obtained as

$$\begin{cases} C_{st0} = 540, C_{st3} = 5, C_{sl0} = 314, C_{sl3} = 4, C_{t0} = 180, C_{t3} = 2, \\ C_{l0} = 67, C_{l3} = 2, C_{m0} = 60, C_{m1} = 50, C_{m2} = 15, C_{m3} = 2 \end{cases} \quad (17)$$

in units of [pF].

4.3. Thrust force characteristics derived from the capacitance matrix

Using the mathematical representation of capacitance variation shown in Eq. (16), the thrust force f_x for the single-frequency method can be mathematically obtained from Eq. (7) as

$$\begin{aligned} f_x(\phi, \theta_x) = \frac{3\pi}{2p} v_0^2 (&C_{m1}\sin(\theta_x - 2\phi + \pi/3) \\ &+ 2C_{m2}\sin(2\theta_x + 2\phi - \pi/3) \\ &- (C_{st3} + C_{sl3} + C_{t3} + C_{l3})\sin(3\theta_x)) \end{aligned} \quad (18)$$

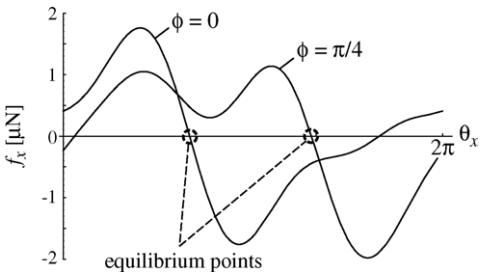


Fig. 12. Intersections of the force characteristics at excitation phases $\phi = 0$, $\pi/4$.

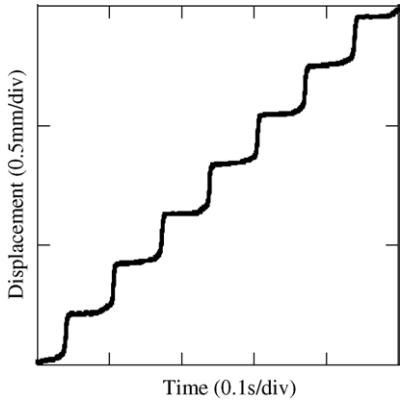


Fig. 13. Slider motion trajectory of the prototype motor driven at 2.5 Hz (v_0 = 1.5 kV).

For the above equation, f_x is plotted in Fig. 11 with the identified parameters in Eq. (17) and a v_0 of 1 V. The intersection of the 3D plot at a constant phase gives the relationship between force and excitation phase. Examples of such intersections are provided in Fig. 12. In the intersectional plots, if the motor has no load, then the points on the plot that intersect the horizontal axis with negative slopes are the equilibrium points for the slider. The bold line in Fig. 11(b), which is referred to as an equilibrium line, is the plot of the equilibrium points calculated for all of the excitation phases, which the slider trajectory should follow when driven at a constant frequency without any load. (Since the plot jumps at some phases, it is connected by dashed lines at these jumps.)

Fig. 13 shows a real slider motion trajectory of the motor, which is driven by three-phase ac voltage of 2.5 Hz (3 mm/s) using single-frequency connection. No load was imposed on the slider, although there is some friction between the slider and the stator. The shape of the trajectory is found to resemble considerably the equilibrium line in Fig. 11(b), verifying the validity of the derived force characteristics.

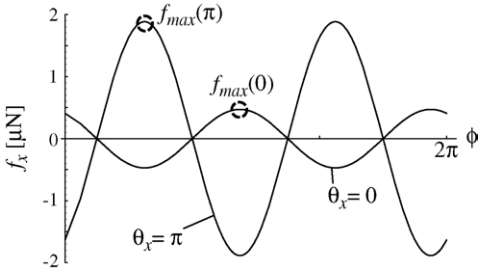


Fig. 14. Intersections of the force characteristics at constant positions, $\theta_x = 0$, π. The thrust force characteristics differ for different slider positions, showing a large thrust force ripple.

5. Improvement of force characteristics based on derived capacitance

5.1. Thrust force ripple

As shown in the previous section, the trajectory of the motor is stepwise, even though it is driven by a sinusoidal wave. This is because the force characteristics differ considerably with position. Fig. 14 shows the intersections of Fig. 11 at constant positions. These plots reveal that the force generation capability differs considerably with slider position; the motor has a large *force ripple*. For practical applications, such a large force ripple is not desirable, because vibrations are generated due to stepwise motion or force control becomes difficult. Therefore, in this section, we introduce and analyze a skew technique by which to reduce the ripple.

5.2. Electrode skew

In magnetic motors, skewing slots are effective for the reduction of force ripple. The same technique can be applied, as electrode skew, to electrostatic motors [12]. Fig. 15 shows an example of skewed electrodes for a DEMED. In this example, the electrodes of the slider are tilted in the driving direction. The electrodes of the slider still have the same pitch p as those of the stator, and their length is l.

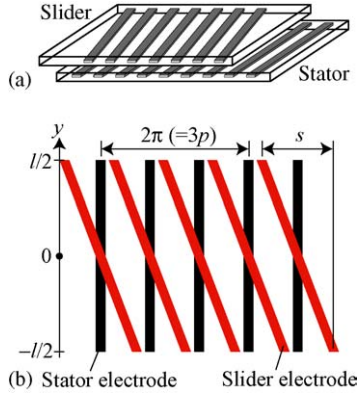


Fig. 15. Electrode skew.

The effect of skew can be analyzed based on Eq. (18). Variable s in Fig. 15 is the *skew factor* in electric angle, which represents the degree of tilt. In skewed electrodes, the relative position between slider electrodes and stator electrodes differs with y. Therefore, the thrust force of an infinitesimal strip at arbitrary y can be expressed as

$$\hat{f}_x(\theta_x, \phi; y)\Delta y = f_x\left(\theta_x + \frac{sy}{l}, \phi\right)\frac{\Delta y}{l} \tag{19}$$

The total thrust force of the skew motor, $f_x(\theta_x, \phi)|_s$, can then be obtained by integrating Eq. (19) as

$$\begin{aligned} f_x(\theta_x, \phi)|_s &= \int_{y=-l/2}^{y=l/2} \hat{f}_x(\theta_x, \phi; y)\,\mathrm{d}y \\ &= \frac{1}{s}\int_{\psi=-s/2}^{\psi=s/2} f_x(\theta_x + \psi, \phi)\,\mathrm{d}\psi \end{aligned} \tag{20}$$

Note that the variable transformation $\psi = sy/l$ is used here.

In order to evaluate the effect of skew in terms of ripple reduction, we define ripple rate r. In a force plot at a constant θ_x, as in Fig. 14, the maximum force can be found, which is referred to herein as $f_{\max}(\theta_x)$. Since $f_{\max}(\theta_x)$ differs with position θ_x because of the ripple, $f_{\max}(\theta_x)$ is maximized or minimized at certain θ_x, which we refer to herein as $\max(f_{\max})$ and $\min(f_{\max})$, respectively. In addition, the average value of $f_{\max}(\theta_x)$, which is referred to as $\mathrm{ave}(f_{\max})$, can be found. Then, the ripple rate r is defined as

$$r = \frac{\max(f_{\max}) - \min(f_{\max})}{\mathrm{ave}(f_{\max})} \tag{21}$$

Plotting the numerically evaluated r with respect to s gives Fig. 16. Ripple rate r cannot be calculated for values of s of 2π and 4π because the thrust force is cancelled out at these skew factors. Therefore, in the plot, the ripple rates for these factors are replaced with that for a normal motor and are indicated with open circles.

The plot shows that force ripple is minimized at $s = \pi$, which is equivalent to $1.5p$. The minimized ripple rate at $s = \pi$ is found to be 0.18, and the maximum ripple rate for a normal motor ($s = 0$) is 1.2, which shows that the ripple is reduced to 15% of the original value. Although, at $s = \pi$,

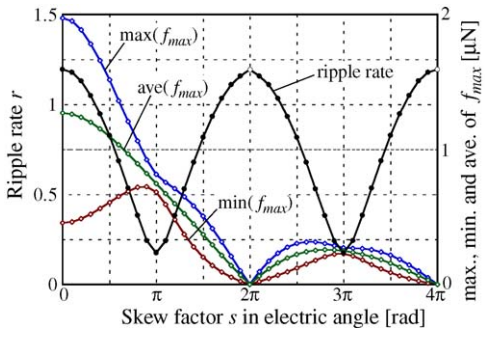


Fig. 16. Effect of skew on reduction of thrust force ripple. Voltage amplitude v_0 for calculation of $f_{\max}$ is normalized to 1 V.

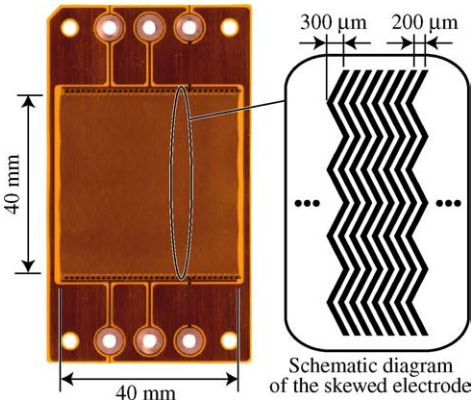


Fig. 17. Slider film with skewed electrode.

ave(f_{max}) is decreased to 58% of the original value, min(f_{max}) is increased to 150%, which, practically speaking, is more useful, as shown later in Section 5.5.

5.3. Parameter identification of a skew motor

Skewed electrodes with a skew factor $s = \pi$ (=1.5p) were fabricated in order to verify the present analysis. Fig. 17 shows a slider film having electrodes that are skewed by 1.5 pitches. The stator film to be combined with the slider is the same as that of the previous prototype. The electrodes are formed into a serrated shape, because skewed electrodes having the tilt shown in Fig. 15 generate a yawing moment on the slider, as can be easily anticipated from their geometry (electrostatic force acts to keep the slider electrodes parallel to the stator electrodes) [12].

Fig. 18 shows the result of capacitance measurement and indicates that the harmonics in capacitance variation are considerably reduced compared to those shown in Fig. 9.

5.4. Simplified motor model

Since the capacitance measured in the previous section has almost no harmonics, this capacitance can be approximated more simply than Eq. (16). Ignoring harmonics, the capacitance matrix for skew motors is approximated as

$$\begin{cases} C_{11}(=C_{22}=C_{33}) = C_{st} \\ C_{44}(=C_{55}=C_{66}) = C_{sl} \\ C_{12}(=C_{13}=C_{23}) = -C_t \\ C_{45}(=C_{46}=C_{56}) = -C_l \\ C_{14}(=C_{25}=C_{36}) = C_{ms}(\theta_x) \\ C_{15}(=C_{26}=C_{34}) = C_{ms}(\theta_x + 2\pi/3) \\ C_{16}(=C_{24}=C_{35}) = C_{ms}(\theta_x - 2\pi/3) \end{cases} \tag{22}$$

where $C_{ms}(\theta_x)$ is defined as $C_{ms}(\theta_x) = -C_0 - C_a \cos(\theta_x)$, and C_{st}, C_{sl}, C_t, C_l, C_0, C_a are all positive constants. Form the measured capacitances, these constants are identified for the

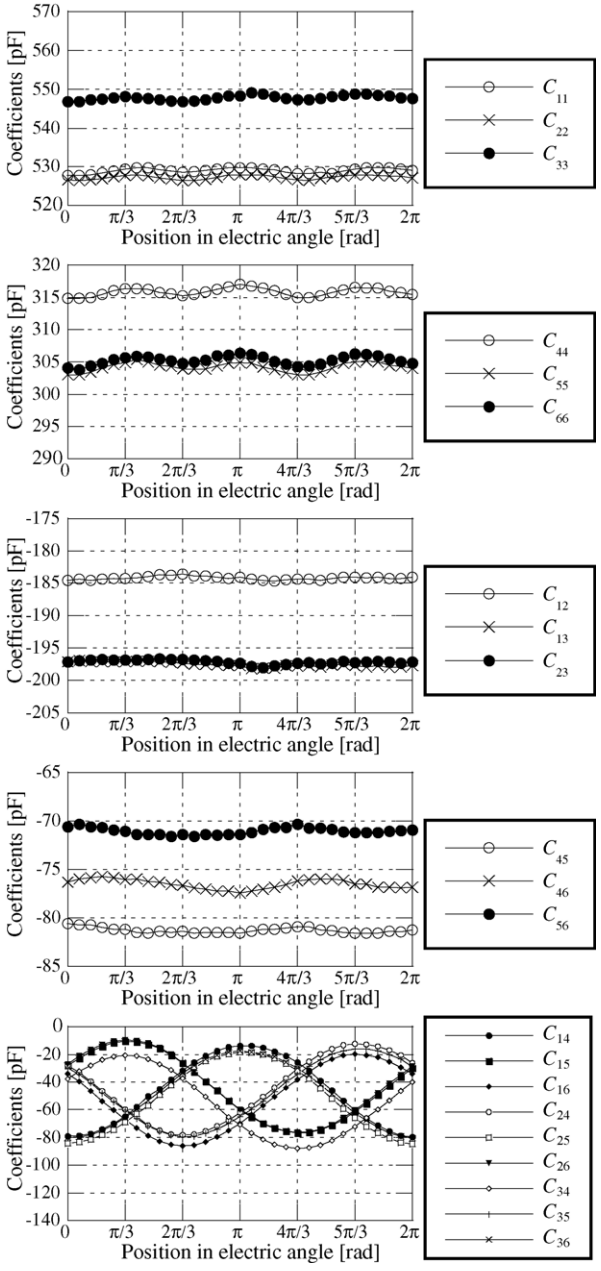


Fig. 18. Measured capacitance coefficient matrix for the skew motor.

prototype as:

$$C_{st} = 535,\ C_{sl} = 308,\ C_t = 193,\ C_l = 76,$$
$$C_0 = 48,\ C_a = 32 \tag{23}$$

in units of [pF]. Substituting Eqs. (4) and (22) into Eq. (7) gives the thrust force of the skew motor as

$$\begin{aligned} f_{xs}(\theta_x, \phi) &= \frac{3\pi}{2p} C_a v_0^2 \sin(\theta_x - 2\phi + \pi/3) \\ &= \frac{3\pi}{2p} C_a v_0^2 \sin(\delta) \end{aligned} \tag{24}$$

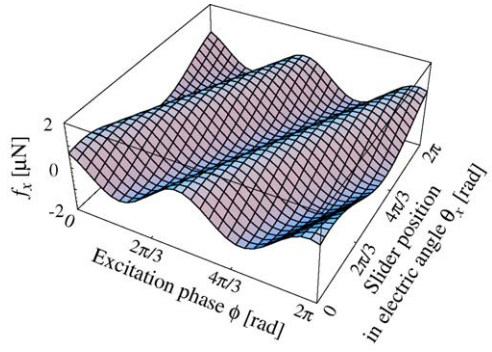


Fig. 19. Thrust force characteristics for the skew motor. Voltage amplitude v_0 is normalized to 1 V.

The force is plotted in Fig. 19 with the parameters shown in Eq. (23) and a v_0 of 1 V. The plot shows that the skew motor has smooth characteristics that do not differ with slider position. Fig. 20 shows the slider motion trajectory of the prototype skew motor, operated under the same conditions as in Fig. 13. Unlike the normal motor, the skew motor shows smooth motion trajectory, from which the effect of skew is verified.

5.5. Quantitative verification of thrust force

Eq. (24) shows that the thrust force is calculated from applied voltage and the geometrical characteristics of the motor. In order to represent the geometrical characteristics with a single coefficient, we define thrust force coefficient κ_t as

$$\kappa_t = \frac{3\pi}{2p} C_a \tag{25}$$

Then, the maximum thrust force at v_0 is given as $\kappa_t v_0^2$ by substituting $\delta = \pi/2$. From the identified value in Eq. (23), the thrust force coefficient κ_t for the skew motor can be obtained as 754 nN/V^2. This value matches the result plotted in Fig. 16, where ave(f_{max}) for $s = \pi$ is 748 nN at $v_0 = 1$ V.

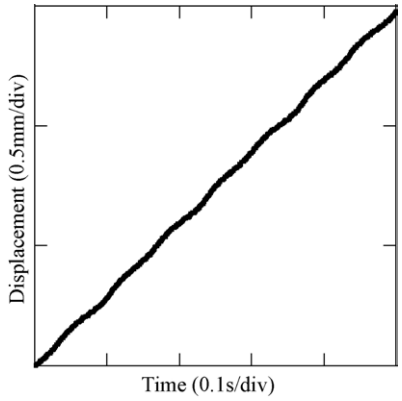


Fig. 20. Slider motion trajectory of the skew motor driven at 2.5 Hz ($v_0 = 1.5$ kV).

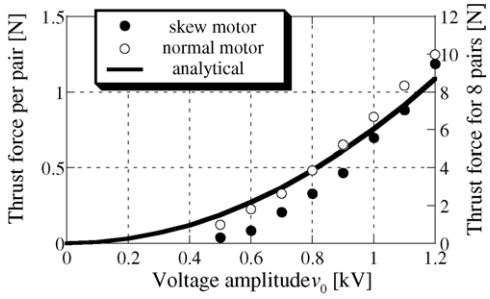


Fig. 21. Analytical and experimental thrust force vs. driving voltage measured at 3 mm/s.

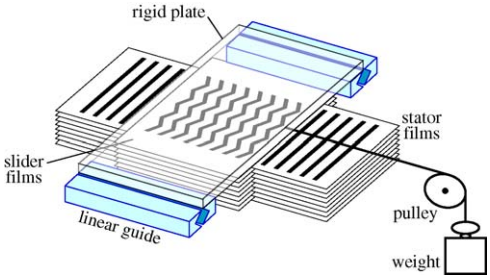


Fig. 22. Schematic diagram of the experimental setup for thrust force measurement.

Fig. 21 plots the variation of the maximum thrust force with respect to voltage amplitude v_0. The line shows the theoretical force, $\kappa_t v_0^2$, and the dots indicate the experimental values, which are determined by measurement using the setup shown in Fig. 22. In this setup, eight pairs of slider and stator films are stacked so as to increase the force generation capability in order to facilitate measurement, and slider motion is constricted only in the thrust direction by using cross-roller linear guides. A weight was then connected to the slider via a wire and a pulley. The maximum weight that could be lifted without stepping-out was determined as the maximum thrust force for each voltage. The intrinsic friction of the linear guides was ignored in the measurements. The slider speed for the experiment was 3 mm/s (2.5 Hz). As a reference, the experimental thrust force for the normal motor was also measured and is also plotted in Fig. 21 using open circles.

Although the experimental values are smaller than their theoretical counterparts at lower voltages, the values generally agree well. The experimental thrust force for the skew motor is approximately equivalent to that for the normal motor, even though the ave(f_{max}) for the normal motor is 70% higher than that for the skew motor, because min(f_{max}) for the normal motor is not larger than that for the skew motor, and thus, if operated at a low speed, such as 3 mm/s, the normal motor steps-out for heavier weights at slider positions at which f_{max} is minimized.

6. Conclusions

The present study describes modeling and parameter identification for a high-power electrostatic motor called the

DEMED. As an electrical model, a six-terminal capacitance network model is described. The thrust force characteristics of the motor can be derived from a capacitance network model. The model parameters were measured by the proposed method using an LCR tester. Based on the measured parameters, the thrust force characteristics were analyzed and compared to the behavior of an actual motor. The comparison revealed good agreement, verifying the validity of the proposed model and measurement method.

Using the proposed model, the effect of skew in the electrostatic motor was analyzed. The analysis showed that a skew of 1.5 electrode pitches, or π in electric angle, can considerably reduce thrust force ripple and can provide ideal thrust force characteristics. The measurements of slider motion trajectory showed that a skew motor demonstrates far smoother motion than a normal motor. Finally, the analyzed force is quantitatively verified experimentally.

In previous research on the electrostatic motor, the performance of the motor was evaluated empirically or numerically, both of which are time-consuming. Using the proposed motor model and parameter identification methods, the motor can be easily evaluated and analyzed, which will facilitate further study on the electrostatic motor for various practical applications. In addition, the method described herein can easily be modified for application to other electrostatic motors and thus contributes to the development of various electrostatic motors, particularly those having high output power.

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Appendix A. Conversion into a three-terminal network model

In the present paper, the motor has been modeled as a six-terminal network. However, for single-frequency operation, the motor model can be simplified into a three-terminal network, which is sometimes more useful.

The simplified voltage vector $\mathbf{v}_s$ for a three-terminal network can be expressed as

$$\mathbf{v}_s = v_0\left[\sin(\phi), \sin\left(\phi - \frac{2\pi}{3}\right), \sin\left(\phi + \frac{2\pi}{3}\right)\right]^T$$

The vector $\mathbf{v}_s$ has the following relationship with vector $\mathbf{v}$ in Eq. (4):

$$\mathbf{v} = \mathbf{T}\mathbf{v}_s$$

where

$$\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}^T$$

Therefore, using the three-phase charge vector $\mathbf{q}_s$, the capacitance matrix $\hat{\mathbf{C}}$ for a three-terminal network can be derived using Eq. (5) as:

$$\mathbf{q}_s = \mathbf{T}^T\mathbf{q} = \mathbf{T}^T\mathbf{C}\mathbf{v} = \mathbf{T}^T\mathbf{C}\mathbf{T}\mathbf{v}_s = \hat{\mathbf{C}}\mathbf{v}_s$$

$$\therefore \hat{\mathbf{C}} = \mathbf{T}^T\mathbf{C}\mathbf{T}$$

Eqs. (7)–(9) also apply to this simplified model.

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